

PRITAM RANJAN KALITA

Project Assistant, Department of Electrical Engineering

Indian Institute of Technology, Jodhpur, India

☎ +91-9101580636 ✉ pritamrk@iitj.ac.in 🔗 LinkedIn 🐙 GitHub

EDUCATION

M.Tech., Robotics Engineering

August 2024 – May 2026

University of Petroleum & Energy Studies, Dehradun, India

- Advisor: Dr. Abhinav Sharma
- Dissertation: “Uncertainty-Aware Trajectory Optimization and Robust Control for Grid-Based Mobile Robots”
- CGPA: 8.9/10

B.Tech., Mechanical Engineering

August 2016 – May 2020

Dibrugarh University Institute of Engineering & Technology, Dibrugarh, India

- CGPA: 7.58/10

EXPERIENCE & INTERNSHIPS

Project Assistant

July 2026 – Present

Wearable Robotics & Control Laboratory, IIT Jodhpur, India

Supervisor: Dr. Saurav Kumar

- Developed technical training documentation on Raspberry Pi 5 and CubeMars AK80-9 V3.0 motor control for onboarding new researchers and supporting prosthetic-leg and exoskeleton development projects in the laboratory.

Robotics and Automation Intern

January 2026 – June 2026

PRD Robotech Solutions Pvt. Ltd., Noida, India

Supervisor: Mr. Prashant Kumar Dwivedi

- Assisted in the development of an obstacle detection and safety feature as part of a Robot Fleet Management Project for preventing collisions between differential-drive robots operating in a 3-way grid-map based warehouse environment using fused data from two 270° 2D LiDAR scans.
- Authored a 3-part technical training manual on the use of ROS 2 Diagnostics package for the company engineers, covering its C++ implementation for monitoring ROS 2 topics, services & actions
- Implemented the A* path planning algorithm (without replanning) from scratch and created a custom Nav2 global planner plugin, replacing the Nav2's default A* global planner plugin and tested it using TurtleBot3, Gazebo Ignition & Cartographer SLAM.

Systems Engineer

April 2022 – August 2023

Infosys Limited, Bhubaneshwar, India

- Worked on frontend development of customer websites using HTML, CSS, JavaScript, & React.

PROJECTS

Uncertainty-Aware Trajectory Optimization and Robust Control for Grid-Based Mobile Robots

January 2026 – May 2026

M.Tech. Thesis Project, UPES Dehradun

Advisor: Dr. Abhinav Sharma

- Developed a Stochastic Nonlinear Model Predictive Control (SNMPC) framework on top of the existing Regulated Pure Pursuit (RPP) controller to optimize its lookahead distance parameter by jointly considering the robot's path tracking error and localization uncertainty instead of relying solely on the velocity-based heuristic of the original RPP algorithm.
- The SNMPC-RPP controller uses a nonlinear path tracking error prediction model, path-relative error covariance propagation and chance constraints for determining an optimal value of the lookahead distance parameter while preserving the rest of the original RPP architecture.
- Validated the controller in ROS 2/Gazebo across multiple operating speeds, achieving reductions of over 40% in cross-track MAE and over 50% in RMSE compared with the baseline RPP controller at 1.5 m/s.

Optimal Control of a Two-Wheeled Self-Balancing Robot Using LQR and Multi-Objective Genetic Algorithm

August 2025 – November 2025

M.Tech. Third-Semester Major Project, UPES Dehradun

Advisor: Dr. Rahul Singhal

- Derived a linearized state-space model for controlling the position and the chassis-tilt angle of a two-wheeled self-balancing (inverted-pendulum) robot, and implemented PID and LQR controllers using MATLAB/Simulink and Simscape Multibody.
- Implemented a Multi-Objective Genetic Algorithm (MOGA) to optimize the LQR weighting matrices Q and R by simultaneously minimizing the Integral Time Absolute Error (ITAE) for position and chassis-tilt angle.
- Compared the performance of the GA-optimized LQR with manually tuned LQR and PID controllers through MATLAB/Simulink simulations, demonstrating faster position convergence while maintaining better chassis-tilt angle stability.

TECHNICAL SKILLS

Programming Languages:	C++, Python, MATLAB
Software & Frameworks:	ROS 2, Nav2, Gazebo, Simulink, Simscape Multibody, SolidWorks
Development & System Tools:	Linux, Git, GitHub, VS Code
Robotics Hardware:	Arduino Uno, Raspberry Pi 5, CubeMars AK80-9 V3.0 Motor
Documentation:	LaTeX, Markdown

RESEARCH INTERESTS

Control Theory, Computer Vision, Sensor Fusion, Robot Localization, Powered Prostheses & Exoskeletons, Medical Robotics, Aerial Robotics

REFERENCES

Dr. Saurav Kumar, PhD

sauravk@iitj.ac.in

+91-94-7171-0777

Assistant Professor

Department of Electrical Engineering

Indian Institute of Technology, Jodhpur

Dr. Abhinav Sharma, PhD

abhinav.sharma@ddn.upes.ac.in

+91-98-9763-6885

Senior Associate Professor

Department of Electrical & Electronics Engineering

University of Petroleum & Energy Studies